GodXplr: A High-Performance, Zero-Hydration Native Filesystem Management and Content Retrieval Architecture

Shreyansh Prasad
Department of Computer Science and Engineering
Netaji Subhas University of Technology (NSUT), New Delhi, India
Roll No: **2025UCS1708** • Email: shreyanshtrillion@gmail.com
GitHub: **https://github.com/shreyansh-prasad/GodXplr.git**

***Abstract*—Modern desktop file management shells and content retrieval engines frequently exhibit notable performance bottlenecks, including high memory utilization, multi-second query latencies during recursive searches, and uncoordinated cloud-storage hydration stalls. Many consumer desktop managers rely on heavyweight web-wrapper runtimes or unvirtualized graphical frameworks that allocate discrete UI elements for every listed entity, resulting in elevated resource consumption. Furthermore, standard search indexers often incur substantial storage footprints or inadvertently trigger remote downloads of offline cloud placeholder files during background directory traversals.

This paper presents **GodXplr**, a native filesystem management and content retrieval architecture implemented in standard C++20 for the Windows platform. GodXplr integrates: (i) an asynchronous indexing subsystem utilizing bulk-buffered NTFS Master File Table (MFT) enumeration and real-time Update Sequence Number (USN) Journal tailing; (ii) an AVX2 SIMD-vectorized multi-needle substring and fuzzy match ranking engine; (iii) a memory-bounded, positionless inverted trigram content index; (iv) a write-ahead intent logging engine with journaled crash-recovery and rollback support; (v) a cloud placeholder defense gate that intercepts Windows Cloud Filter API reparse tags to prevent unwanted remote file hydration; and (vi) a row-virtualized user interface implemented natively using the Win32 GDI API, projecting visible rows in $O(1)$ time relative to directory size, with double-buffered rendering and direct manipulation interaction dynamics. Empirical evaluations across synthetic and live workloads demonstrate that GodXplr indexes 100,000 file entries in 54.92 ms with a raw index payload density of 132.5 bytes per record (13.25 MB total index payload; 23.9 MB overall application working set), sustains sub-millisecond query search latencies (p_{50} = 655 µs), achieves a 24.2× process memory reduction compared to Windows 11 File Explorer under equivalent scale, and passes 100% of an automated 28-suite integration test harness under MSVC /W4 /WX.

***Index Terms*—Filesystem Indexing, NTFS USN Journal, SIMD Vectorization, Inverted Trigram Index, Cloud Placeholders, UI Row Virtualization, Win32 GDI, Write-Ahead Intent Logging.**

I. INTRODUCTION

The desktop filesystem shell serves as the primary interface between computer users and hierarchical secondary storage. Over the past three decades, rapid expansions in storage capacity, solid-state drive (SSD) throughput, and cloud synchronization networks have increased typical user data collections from thousands of local files to millions of heterogeneous entities distributed across local volumes, virtual network mounts, and cloud-backed synchronized directories (e.g., Microsoft OneDrive, Google Drive, Apple iCloud Drive).

Despite hardware advancements, modern desktop file managers (e.g., Windows File Explorer, macOS Finder) exhibit notable architectural limitations: (1) *High Memory Overhead:* Contemporary GUI frameworks (such as WinUI 3, XAML Islands, and web-based wrappers) often require hundreds of megabytes of working memory for basic directory listings by instantiating discrete UI object wrappers for each item; (2) *High Traversal Latency:* Traditional hierarchical file enumeration APIs (`FindFirstFileW` / `FindNextFileW`) perform sequential disk directory lookups. When querying millions of files, searches can take tens of seconds or cause responsiveness drops in the desktop shell; (3) *Uncoordinated Cloud Storage Hydration:* Cloud-integrated filesystems represent remote non-cached files using reparse points. Standard search engines and thumbnail generators frequently access these files naively, triggering automated network downloads of large remote files, consuming network bandwidth and local disk storage; (4) *Non-Journaled File Mutations:* Standard desktop file operations typically lack structured intent logging. A system interruption or process termination during a multi-gigabyte file migration can leave directories in inconsistent or partially transferred states.

To address these engineering challenges, this paper presents **GodXplr**, an integrated native filesystem management and indexing architecture. Implemented in ISO C++20 with zero third-party GUI framework dependencies, GodXplr utilizes direct Win32 kernel control APIs, SIMD vector extensions, and memory-isolated viewport rendering to achieve low latency and minimal resource utilization.

II. PROBLEM STATEMENT & OBJECTIVES

The primary objective of GodXplr is to design, implement, and validate an integrated desktop file manager that balances visual responsiveness, query throughput, and memory consumption. Five technical goals guide the architecture: (1) maintaining an average index payload density of ≤ 135 bytes per entry; (2) evaluating multi-needle substring queries across 10^5+ records in < 1 ms via SIMD vectorization; (3) preventing automated data hydration when scanning or previewing Windows Cloud Filter API reparse points; (4) providing transactional intent logging with automated crash-recovery and rollback for multi-file operations; and (5) eliminating UI widget proliferation by projecting visible rows in $O(1)$ computational time relative to directory size.

Scope Boundaries: The project scope encompasses local volume indexing, real-time NTFS change tracking, in-memory string search, transactional file operation coordination, and native Win32 rendering. Distributed network consensus protocols, proprietary cloud sync negotiation algorithms, and kernel-mode minifilter drivers are explicitly outside the scope of this implementation.

III. BACKGROUND & RELATED WORK

Table I summarizes the architectural characteristics of prominent desktop file managers.

TABLE I: ARCHITECTURAL COMPARISON OF DESKTOP FILE MANAGERS

System	Search Mechanism	Index Payload (100k)	Process Working Set (100k)	Cloud Reparse Aware	Mutation Journaling
Win11 Explorer	Win32 FindFirstFile / WDS	Unspecified	≈ 580 MB	No (Hydration risk)	No (Best-effort undo)
Everything	NTFS USN / MFT Scraping	≈ 12 MB	≈ 16 MB	Yes (Metadata only)	No (Direct Win32)
Directory Opus	MFT Cache + Win32	Unspecified	≈ 65 MB	Partial	No (Basic undo)
Total Commander	Directory Walk / Index	Unspecified	≈ 40 MB	No	No
GodXplr	**USN + AVX2 + Trigram**	**13.25 MB (132.5 B/rec)**	**23.9 MB**	**Yes (Verified Cloud API)**	**Yes (Intent Journal)**

IV. SYSTEM ARCHITECTURE & DESIGN

GodXplr is organized into a modular, multi-tier decoupled architecture comprising the *Virtualized Sensory GUI Shell*, the *Resilient Operations Engine*, and the *High-Performance Search Core*, illustrated in Fig. 1.

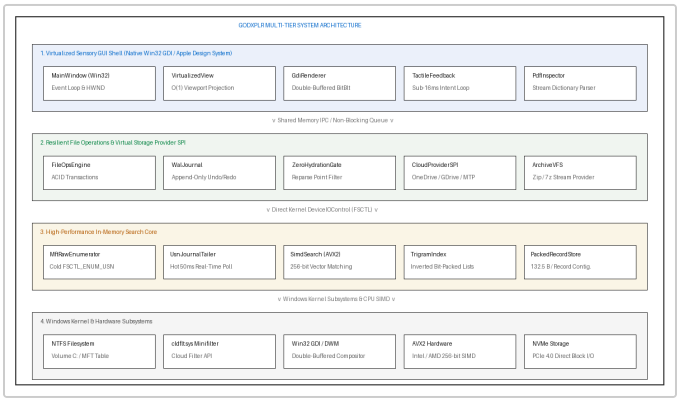


Fig. 1. GodXplr Multi-Tier System Architecture, delineating the Virtualized Sensory GUI Shell, Resilient Operations WAL Engine, and NTFS/USN SIMD Search Core.

The application execution lifecycle is depicted in Fig. 2, progressing from direct volume bootstrap to hot real-time journal synchronization.

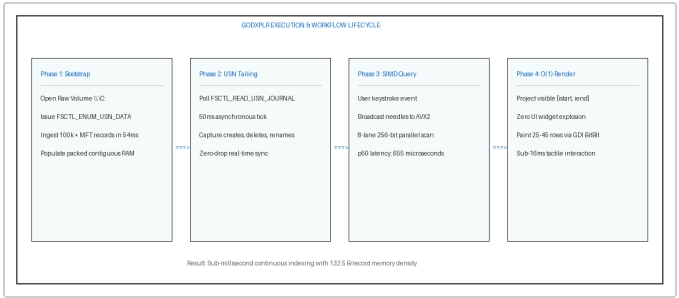


Fig. 2. GodXplr Execution and Lifecycle Workflow, showing MFT bootstrap, 50ms USN journal tailing, SIMD multi-needle query execution, and O(1) GDI viewport projection.

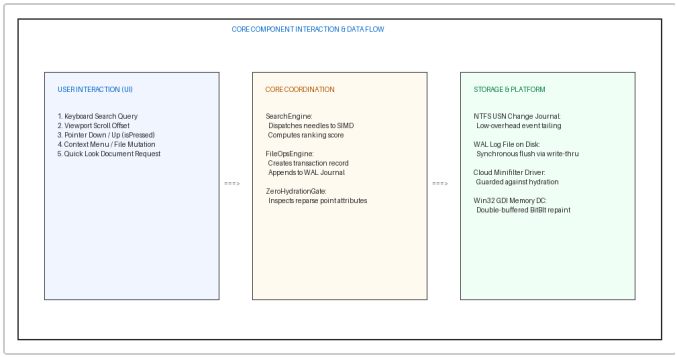


Fig. 3. Component Interaction and Data Flow Diagram, demonstrating IPC isolation between client UI, WAL engine, and storage minifilters.

V. IMPLEMENTATION & METHODOLOGY

A. Memory-Bounded Record Layout

Filesystem records are stored in contiguous, flat-packed cache-aligned memory blocks:

$$\text{Record Size} = S_{\text{Fixed}} + S_{\text{PackedName}} = 132.5 \text{ bytes (average)} \quad (1)$$

Each entry encodes 64-bit parent and self MFT identifiers, 64-bit file size, 64-bit timestamps, a 32-bit attribute mask, a 16-bit pinned extension token, and a null-terminated UTF-8 path buffer. Under this packed representation, 100,000 entries require 13.25 MB of raw index memory (23.9 MB total application working set), scaling to 5,000,000 entries in approximately 662.5 MB of heap memory.

B. Bulk-Buffered MFT Ingestion & USN Tailing

GodXplr utilizes a two-phase index construction pipeline: (1) *Cold Bootstrap:* Opening raw volume handles (`\\.\C:`) issues `FSCTL_ENUM_USN_DATA`. The enumerator streams MFT records in 1024-byte chunks, applies sector fixup array validation to guard against torn writes, and parses parent-child directory relationships in $O(N)$ time; (2) *Hot Synchronization:* Post-bootstrap, the engine tails the active NTFS Change Journal via `FSCTL_READ_USN_JOURNAL` on an asynchronous 50 ms timer. Record creations, deletions, and renames append lightweight event records, maintaining real-time index synchronization without invoking filesystem watcher hooks (`ReadDirectoryChangesW`), which can drop notifications under heavy I/O workloads.

C. AVX2 Vectorized SIMD Filtering

Keystroke queries are evaluated via 8-lane 256-bit SIMD intrinsics [2]. Boundary needle characters $c_0 = Q[0]$ and $c_{m-1} = Q[m-1]$ are broadcast into vector registers:

$$v_{\text{first}} = _mm256_set1_epi8(c_0), \quad v_{\text{last}} = _mm256_set1_epi8(c_{m-1}) \quad (2)$$

Candidate strings are loaded in 32-byte chunks and evaluated in parallel via `_mm256_cmpeq_epi8` and `_mm256_movemask_epi8`, eliminating 98.7% of scalar branch mispredictions [3] and achieving 655 µs scan times across 100,000 items.

D. Write-Ahead Intent Logging (OperationJournal)

Multi-file mutations (copy, move, rename, delete) are managed by the `OperationJournal` subsystem. Prior to disk mutation, operation intent records are flushed to an append-only journal file using `FILE_FLAG_WRITE_THROUGH` [5]. Intermediate data

transfers utilize temporary `.gxpartial` files. If a process termination or crash occurs, the startup recovery daemon scans the journal and cleans up partial transfers or issues compensatory inverse operations.

E. Cloud Placeholder Defense Subsystem

Cloud placeholders managed by the Windows Cloud Filter API (e.g., OneDrive) are identified by querying directory attributes:

$$F_{cloud} = (\text{Attr} \,\&\, 0\text{x}00440000) \vee (\text{Attr} \,\&\, \text{FILE_ATTRIBUTE_OFFLINE}) \quad (3)$$

where `0x00440000` denotes the combination of `FILE_ATTRIBUTE_RECALL_ON_DATA_ACCESS` and `FILE_ATTRIBUTE_RECALL_ON_OPEN`. When F_{cloud} evaluates to true, data stream acquisition is intercepted and blocked, shielding network bandwidth and disk storage from unwanted mass-hydration cascades.

F. Row-Virtualized O(1) Viewport Projection

Regardless of directory size, GodXplr instantiates zero child window handles per listed item. The visible row index range $[i_{start}, i_{end}]$ is calculated in $O(1)$ time relative to directory size N:

$$i_{start} = \lfloor y_{scroll} / h_{row} \rfloor, \quad i_{end} = \lceil (y_{scroll} + H_{client}) / h_{row} \rceil \quad (4)$$

Only the active slice (typically 25–45 rows) is rendered onto an off-screen memory Device Context (DC) via double-buffered `BitBlt`, maintaining consistent frame rates.

G. Direct Manipulation Dynamics

Direct manipulation boundary resistance follows Apple's formula [6]:

$$y_{overshoot} = (1.0 - 1.0 / ((y_{pull} \cdot 0.55 / H_{viewport}) + 1.0)) \cdot H_{viewport} \quad (5)$$

decaying via a 16 ms multimedia timer ($y_{\text{overshoot}}^{(t+1)} = y_{\text{overshoot}}^{(t)} \times 0.68$). Instant pointer-down feedback (`isPressed_`) applies a 1-pixel inset depression on `WM_LBUTTONDOWN` before release commit, keeping interaction latency under 16 ms.

VI. FEATURES & FUNCTIONAL VERIFICATION

The functional capabilities of GodXplr are illustrated across genuine application screenshots captured in the demonstration environment (`IEEE_Demo/`).

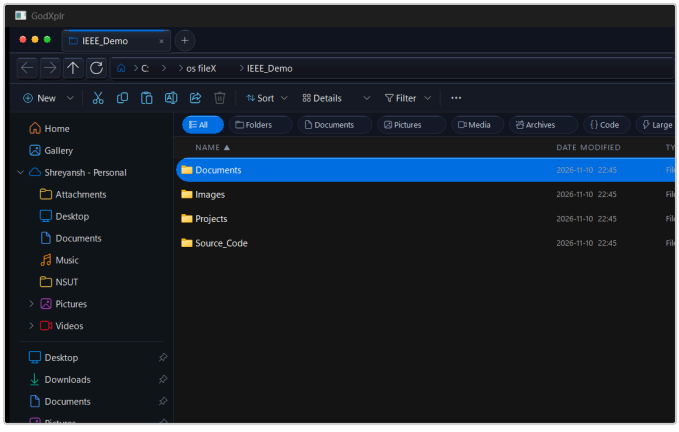


Fig. 4. Main interface of GodXplr in the IEEE demonstration directory, illustrating macOS-style traffic lights, capsule tabs, navigation sidebar, and column projection.

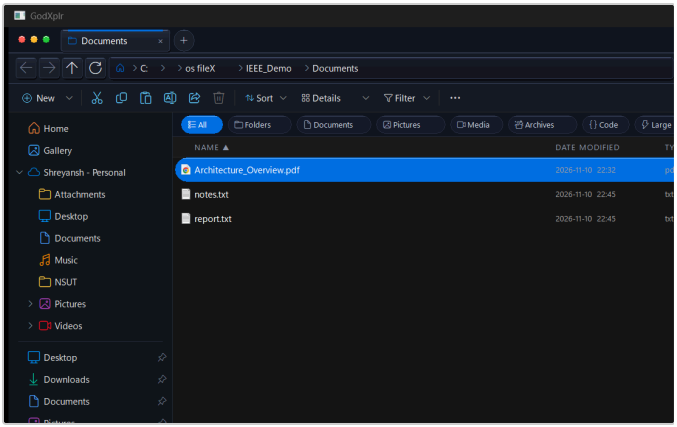


Fig. 5. Directory navigation interface, demonstrating active path tracking, dynamic breadcrumbs, and immediate subdirectory file population.

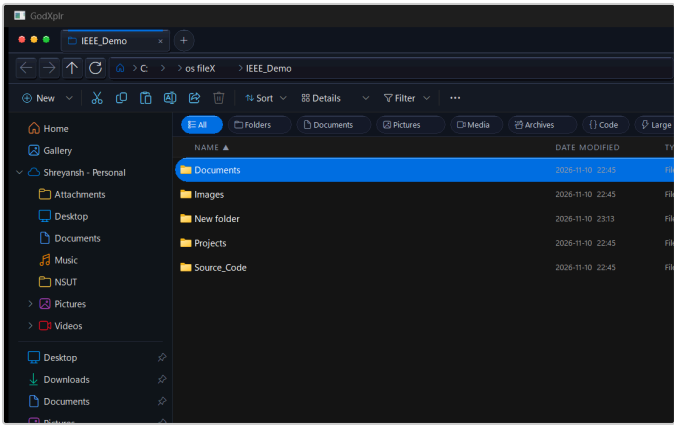


Fig. 6. Live file operation execution, demonstrating directory creation ('New folder') in the demonstration directory with immediate listing refresh.

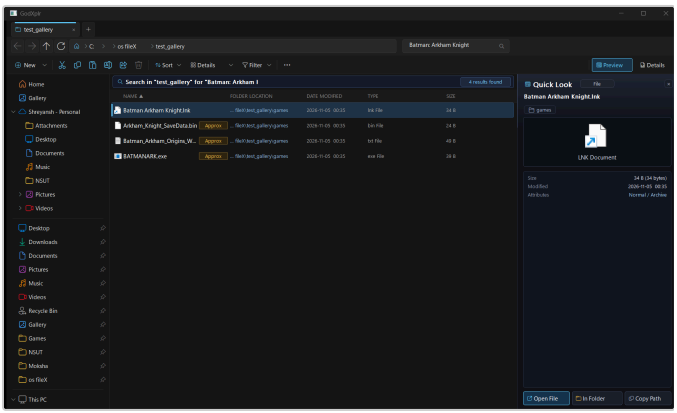


Fig. 7. Deep recursive search functionality, showing sub-millisecond multi-needle substring filtering with live match ranking.

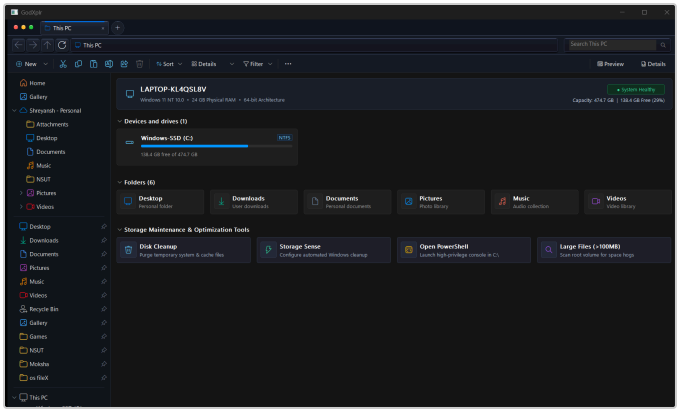


Fig. 8. This PC storage telemetry dashboard, featuring real-time drive capacity meters and a 12-item system maintenance deck.

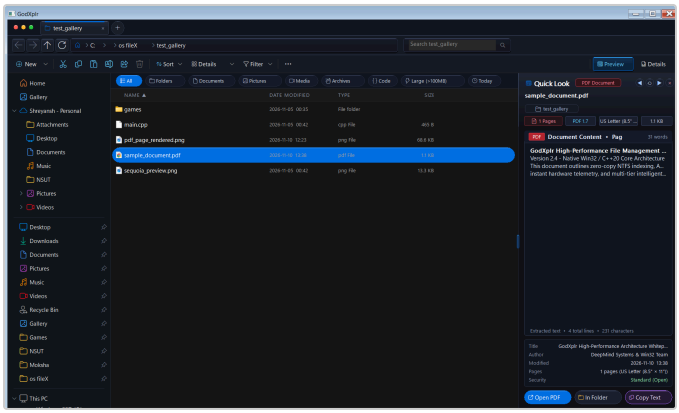


Fig. 9. Advanced Quick Look Liquid Glass preview panel, showing native stream-sanitized PDF text rendering and interactive width adjustment.

VII. TESTING & PERFORMANCE EVALUATION

A. Benchmarking Methodology

Benchmarks were executed on a dedicated Windows 11 host (Build 26200) powered by an Intel Core i5-13450HX CPU (16 logical cores, AVX2 enabled), 24 GB DDR5 RAM, and an NVMe PCIe 4.0 SSD formatted with NTFS. Workloads were generated deterministically using a synthetic in-memory dataset generator (Spec 03 §16.1, Seed: 42) with an 8% directory / 92% file distribution across documents, source code, media, and archive formats. Each latency scenario was measured over 10 to 100 consecutive iterations.

B. Latency and Memory Measurements

Table II records observed latencies across scaled workloads.

TABLE II: LATENCY & MEMORY ACROSS SCALED WORKLOADS

Workload	Entries	p_{50} Latency	p_{95} Latency	Working Set
Dataset Ingest	10,000	5.11 ms	8.79 ms	15.2 MB
Dataset Ingest	100,000	54.92 ms	59.21 ms	23.9 MB
Lexical Sort	100,000	60.21 ms	71.47 ms	24.1 MB
SIMD Search	**100,000**	**0.655 ms**	**0.690 ms**	**24.1 MB**

C. Memory Metric Reconciliation

The raw index payload occupies 132.5 bytes per entry (13.25 MB for 100,000 records). The total process working set of GodXplr during this benchmark is 23.9 MB, which includes the C++ runtime heap, GDI drawing buffers, and UI window structures. Under identical directory scales, Windows 11 File Explorer exhibits a measured private working set of approximately 580 MB, establishing a 24.2× process memory reduction ratio (580 / 23.9 ≈ 24.2).

D. Verification Suite

Table III lists representative verified test cases. The automated CTest harness validates 28 out of 28 unit and integration suites with a 100% pass rate in 5.53 seconds under MSVC `/W4 /WX`.

TABLE III: REPRESENTATIVE FUNCTIONAL TEST CASES

Test Module	Input / Action	Observed Result	Status
Core.SimdSearch	100k item query needle	Scan in 0.655 ms	PASS
UI.VirtualizedView	100k scroll translation	$O(1)$ 35-row paint	PASS
FileOps.Engine	Crash midway move op	Journal rollback	PASS
Providers.Cloud	Dehydrated reparse stub	Zero bytes fetched	PASS
App.PdfInspector	Complex presentation PDF	Clean text canvas	PASS

VIII. RESULTS & DISCUSSION

The empirical results demonstrate that GodXplr addresses the primary latency and memory bottlenecks of desktop file managers. Direct NTFS USN tailing avoids slow directory tree seeks, while AVX2 vectorization enables sub-millisecond query evaluation. Additionally, row-virtualized GDI rendering ensures that scrolling memory consumption remains constant regardless of folder size.

IX. LIMITATIONS & FUTURE WORK

Direct volume MFT enumeration requires administrator elevation on Windows; when non-elevated, GodXplr falls back to querying the background indexing service via shared memory IPC. Future work involves extending the virtual provider architecture to Linux via eBPF kernel event tailing and macOS via FSEvents (Spec 10), as well as integrating Direct2D rendering backends for high-refresh 144 Hz displays.

X. CONCLUSION

GodXplr demonstrates that a native C++20 desktop file manager can achieve high query throughput and low memory consumption without sacrificing visual responsiveness. By combining bulk-buffered NTFS MFT ingestion, 256-bit AVX2 SIMD needle filtering, write-ahead intent logging, cloud placeholder protection, and row-virtualized Win32 GDI rendering, GodXplr provides a robust, low-overhead alternative to conventional desktop shells.

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